

AMLH 2025-2026 Assignment Instruction

Deadline for submission: Wednesday, 08 July 2026, 16:00 BST

•Length: The summative assessment refers to the outcome of the entire module and culminates in a documented paper of 2000-2500 words (excluding figures, references) and the submission of an iPython-notebook.

•Python packages: tensorflow, pytorch, keras, scikit-learn, etc.

•Each student needs to select one dataset as your choice from the provided list. The maximum number of students for any topic/dataset is 54. First come first served. (Last year, all students selected their favorite datasets)

Dataset options for the Assignment:

- A. Lateral flow test classification**
- B. Glucose prediction for diabetes**
- C. NLP- patient question classification**

The assignment should follow this overall outline (the requirements of each dataset are provided in a separate section):

1. Introduction		Words:
	Please describe the background, motivation and importance of the data in light of related literature. Show a sound interpretation of the medical problems presented in the data, and introduce necessary concepts and existing research. [10 marks]	350

2. Methodology		
2.1 Preprocessing		Words:
	Outline the selected dataset (including features and class labels) and provide descriptive statistics of the contained variables. Visualise the data or feature space in a plot if it is possible and explore the underlying characteristics. Please describe details of preprocessing of the data, including data cleaning, imputation, normalization, augmentation, up or down sampling, feature engineering etc. of the chosen dataset. [15 marks]	450
2.2 Algorithm design and implementation		Words:
	Select two AI models of the course, or two neural networks which have very different structures. Given a high-level description of both algorithms including their rationale or model structure. Describe and demonstrate for both algorithms:	700

	a) How to generate training and testing data with appropriate format that suit the AI model b) Optimization of hyper-parameters c) Model evaluation based on the outcomes including widely-used criteria that mentioned in the course Demonstrate your solution with an attached iPython notebook. Ensure reproducibility and transparency. [40 marks]	
3. Results		Words:
	Present optimized hyper-parameters and reasonable evaluation criteria such as a confusion matrix, precision, recall, RMSE, MARD, F1, AUC and a ROC-plot, etc. Provide an analysis for both algorithms with different parameters and give a textual description of the results. (please choose appropriate metrics carefully) [25 marks]	600
4. Discussion and Conclusion		Words:
	Compare and discuss your findings (results of two algorithms). If it is possible, it would be good to compare with other scientific publications that used the same medical dataset. Discuss how you would improve your methodology, current limitations and future work. [10 marks]	400
5. Reference		
6. Appendix		
	Attach a reproducible iPython Jupiter notebook.	
Total: 100 marks		

Plagiarism or collusion is not allowed. The module can be failed straightaway if the assignment or codes notebook fails in the plagiarism test (please note the assignment is within AI category 2).

Markers will look for the following sections in the assignment:

- Sound understanding of the provided dataset and appropriate pre-processing to obtain a dataset that is suitable for the machine learning model
- Appropriate selection and learning/training of two algorithms (one of them can be seen as the baseline) to address the target problems in medical imaging, time series or NLP
- Evaluation of the performance and meaningful discussion
- Other requirements that have been asked in the associated dataset instruction
- The layout, presentation, references of the paper/report

If there is any question regarding the dataset or coursework, please post it in the forum on moodle page, or email Kevin/Ken/Julia directly.

Provisional mark and feedback will be released in Autumn 2026.

AMLH 2026 - NLP Coursework: Patient Question Classification

1. Introduction

In this assignment, you are required to build a patient question classification system using Natural Language Processing (NLP) techniques. You will be able to predict the most likely disease or condition associated with a patient question.

Patient question classification is an important component of many healthcare AI systems. Correctly identifying the underlying disease from a patient's question can support information retrieval, patient triage, and clinical decision support systems. The objective of this coursework is comparing traditional NLP approaches and modern neural approaches (such as Large Language Models (LLMs)) for the task.

2. Dataset

a. Description

The dataset is derived from the patient information section of the NHS.UK website, which provides explanations of various diseases, their symptoms, and treatments. It was synthetically generated using ChatGPT and released as part of the OpenGPT dataset by the CogStack KCL/UCL team (<https://github.com/CogStack/opengpt>). The test set has been manually validated by a human expert to ensure accuracy. This methodology is state-of-the-art in NLP [1,2].

The dataset covers a wide range of medical conditions (e.g., bronchiolitis, laryngitis, acne, gallstones, pneumonia, tonsillitis) and includes multiple Q&A pairs for each condition. It consists of:

- Two CSV files containing patient questions and corresponding disease labels (for training and testing).
- A ZIP archive (db_nhs_qa_classification.zip) with 906 plain text documents from the NHS.UK website. Each document contains information relevant to answering questions about a specific disease or condition (e.g., face blindness).

b. Data format

CSV Files:

The dataset includes two CSV files:

- **patient_qa_classification_train.csv** – the training set, containing approximately 8,900 questions and corresponding disease labels across 906 diseases.
- **patient_qa_classification_test.csv** – the test set, with 200 question-disease pairs across 102 diseases. The 102 diseases appearing in the test set are also present in the training set.

Each file contains the following columns (see the table below):

- **question**: a natural language question related to a medical condition
- **disease**: the associated disease or condition
- **answer**: a reference answer to the question
- **reference_url**: the NHS.UK webpage from which the document for the zip archive was scraped.

question	disease	answer	reference_url
How is IVA diagnosed?	isovaleric_acidaemia	Newborn blood spot screening is offered to check if babies have IVA, involving pricking the baby's heel to collect drops of blood for testing.	https://www.nhs.uk/conditions/isovaleric-acidaemia/
How can one be exposed to asbestos?	asbestosis	One can be exposed to asbestos by inhaling its fine dust that contains asbestos fibres, which can be released if material containing asbestos is damaged.	https://www.nhs.uk/conditions/asbestosis/
What is a stoma nurse?	colostomy	A stoma nurse is a specialized nurse who has received training in caring for colostomy patients. They offer support, advice and practical guidance on how to look after the stoma and how to prevent any infection or irritation that may occur.	https://www.nhs.uk/conditions/colostomy/

For the plain text files: Each file is named in the format of xxx.txt, where xxx is a disease. The disease value appearing in the CSV disease column corresponds directly to the filename after removing the .txt extension: for example, *gallstones* corresponds to *gallstones.txt*

The plain text documents may be used as an external knowledge source for retrieval-based or retrieval-augmented approaches.

Acknowledgments: The dataset is public and is available at https://github.com/CogStack/OpenGPT/blob/main/data/nhs_uk_full/prepared_generated_data_for_nhs_uk_qa.csv

3. Problem Formulation

The primary objective of this coursework is to build a patient question classification system that can accurately identify the disease associated with a patient question. Given a patient question, the system should assign the most appropriate disease label from the set of 906 disease classes available in the training data. This is therefore a multi-class text classification problem with 906 possible output labels.

For example:

Question

Is there a specific test for prosopagnosia?

Predicted Disease

face_blindness

Please refer to the practical session in Week 10 - Building a Retrieval-Augmented Generation (RAG) System Using Large Language Models (LLMs) for Patient Q&A for sample codes.

4. Requirements on coursework report:

The report typically is composed of the following sections.

Introduction

This section provides the background and literature review for the coursework. Discuss why patient question classification is an important healthcare NLP task, and what NLP techniques have been used to solve similar problems in recent years. Provide an overview of the approaches investigated in this coursework and how they relate to recent literature.

Preprocessing:

This section describes and justifies all preprocessing and conversion steps applied to the data.

Where relevant, justify preprocessing choices such as tokenisation, lowercasing, stop-word removal and lemmatisation. Text conversion may be performed using TF–IDF representations, for example through the `TfidfVectorizer` class from the `scikit-learn` (`sklearn`) library, or using transformer tokenisers such as the `AutoTokenizer` class from the `transformers` library. If you use these classes, describe the preprocessing and conversion settings used. If default settings were not modified, look up and report the relevant default behaviour.

Report preprocessing and text conversion settings in a dedicated table.

Methodology

Implement and compare at least one traditional NLP approach and one neural-based approach, such as LLMs. Include diagrams to illustrate the workflow of each method. A suitable traditional baseline is a TF–IDF-based information retrieval system. In this approach, patient questions are represented as TF–IDF vectors, and disease labels for test questions are predicted using the labels associated with the most similar training questions. Describe and justify important hyperparameter choices, such as the number of retrieved neighbours and similarity thresholds.

The neural approach may use an LLM. Possible approaches include zero-shot prompting, few-shot prompting [3], or fine-tuning a BERT-based classifier. Examples are provided in Labs 9 and 10.

Describe the complete experimental protocol in sufficient detail to enable reproduction of the results. This should include key hyperparameters and prompts (reported in a dedicated table), computational resources used, and any ablation studies or model comparisons performed. If a validation set is used, describe how it was constructed. Example hyperparameters include temperature, maximum number of generated tokens, learning rate, batch size, and number of epochs. Briefly justify the main design choices.

Results analysis

Evaluate model performance on the test set using accuracy, defined as:

$$\text{Accuracy} = \frac{\text{Number of Correctly Labelled Samples}}{\text{Total Number of Samples}}$$

Additional metrics may also be reported together with a brief explanation of the additional insights they provide beyond accuracy.

Present results using tables and diagrams wherever possible. Where model training is performed, include training and validation loss curves. A detailed error analysis is expected, including examples of correct and incorrect predictions per method.

Discussion

Discuss the relative strengths and weaknesses of the traditional and neural (LLM-based) approaches, considering factors such as predictive performance, computational requirements and ease of implementation. Reflect on how methodological choices, such as preprocessing, training settings, and prompt design may influence performance. The discussion should also consider the broader implications of patient question classification systems in healthcare.

Organisation of Report

Introduction

1.1 Literature Review

1.2 Motivation and Rationale

Methodology

2.1 Dataset Description

2.2 Data Preprocessing

2.3 Traditional NLP Approach

2.4 Neural (LLM) Approach

2.5 Experimental Protocol

Results

3.1 Exploratory Data Analysis

3.2 Performance Comparison

3.3 Error Analysis

Discussion

4.1 Comparison of Approaches

4.2 Impact of Design Choices

4.3 Implications for Healthcare

5. Coding

Demonstrate your solution with an attached iPython notebook. Ensure reproducibility and transparency.

References

[1] Dubois, Y. et al. (2023). AlpacaFarm: A simulation framework for methods that learn from human feedback. *Neural Information Processing Systems*, abs/2305.14387, pp.30039–30069.

[2] Kweon, S. et al. (2024). Publicly shareable clinical large language model built on synthetic clinical notes. In: *Findings of the Association for Computational Linguistics ACL 2024*. 2024.

[3] Schulhoff, S. et al. (2024). The prompt report: A systematic survey of prompt engineering techniques. *arXiv [cs.CL]*.